ThermoX POWER BOARD: COPPER (60 x 60 mm square)

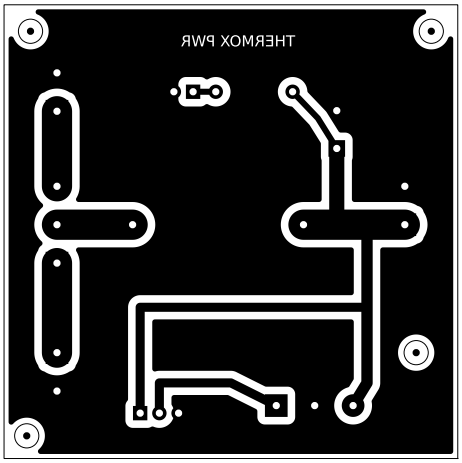
Print at 100 % / Actual size (no 'fit to page'). Check the 50 mm bar with a ruler.

Toner transfer or UV film: use this page as it is (do not mirror). Copper is on the bottom of the board.

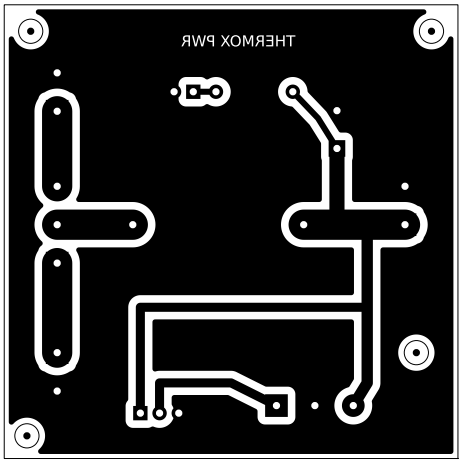
Check: the small text 'THERMOX PWR' looks mirrored on the paper and reads normally on the etched copper side.

Drill: 1.3 mm screw terminals, 1.5 mm fuse holder, 1.0 mm capacitor and J7 header, 0.8 mm R1/LED, 3.2 mm mounting holes.

The wide tracks carry up to 7 A: cover them with a layer of solder after etching.



copy 1



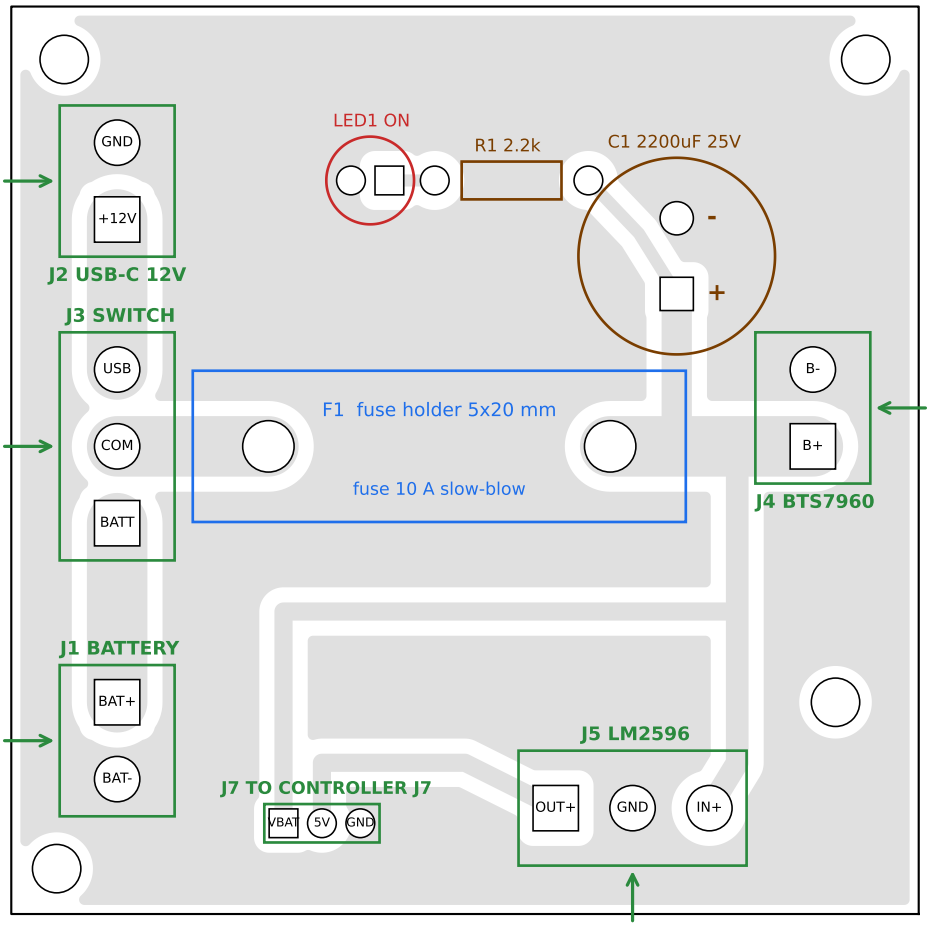
copy 2 (spare)

50 mm: measure this after printing. It must be exactly 50 mm.

ThermoX POWER BOARD: PARTS SIDE (top view, enlarged 2 x)

Parts go on this side, copper is underneath (grey). Square pad = pin 1 (and + on C1, anode on LED1).

Arrows show where the wires go into the screw terminals. C1: the stripe (-) of the capacitor towards the '-' pad.



ThermoX POWER BOARD: WIRING, PARTS, NOTES

CONNECTORS

J1 BATTERY	BAT+: battery + after the BMS (P+). BAT-: battery - (BMS P-).
J2 USB-C 12V	+12V and GND from the PD trigger board output. Set the trigger board to 12 V and measure it first.
J3 SWITCH	ON-OFF-ON toggle switch, 10 A DC or more. Middle lug to COM, one outer lug to BATT, the other to USB. Middle position = everything off.
J4 BTS7960	B+ and B- to the BTS7960 power terminals (thick wire, 1.5 mm2). M+/M- of the BTS7960 go to the Peltier.
J5 LM2596	IN+ to LM2596 IN+, GND to LM2596 IN- (IN- and OUT- are the same on the module), OUT+ from LM2596 OUT+. Set the LM2596 to 5.0 V before connecting it.
J7 TO CONTROLLER	VBAT, 5V, GND to J7 VBAT, 5V, GND on the controller board (pin 1 = VBAT on both, square pad).

IMPORTANT

Only ONE source at a time: the switch picks battery or USB-C. Never connect the trigger board straight to the battery.
The trigger board does not charge the battery; on USB-C the battery is just not used.
Peltier power on USB-C (phone page slider): 12V 5A charger and 5A cable -> 60 % max; 12V 3A -> 40 % max.
Never set the trigger board to 15 V or 20 V: the 12 V fan and the TEC1-12706 would be damaged.
Fuse F1 (10 A slow-blow, 5x20 mm) protects both sources. C1 smooths the Peltier PWM so the charger does not cut out.
Before the first power-up: check with a meter that BAT+, +12V, COM and BUS are not shorted to GND. Then, with the switch on BATT, LED1 lights and J7 VBAT reads the battery voltage. Switch to the middle (off) before changing source.
Cover the wide copper tracks with solder after etching (they carry up to 7 A).

PARTS

J1, J2, J4: 2-pin screw terminal 5.08 mm (KF301-2P) J3, J5: 3-pin screw terminal 5.08 mm (KF301-3P)
J7: 1x3 male header 2.54 mm + 3 dupont wires to the controller J7
F1: 5x20 mm PCB fuse holder (BLX-A, pins 22.6 mm apart) + 10 A slow-blow fuse
C1: 2200 uF 25 V electrolytic, up to 13 mm diameter, 5 mm lead spacing
LED1: 3 or 5 mm LED, R1: 2.2 kOhm 1/4 W, SW: ON-OFF-ON toggle 10 A (on the case, wired to J3)
M3 screws/standoffs: 4 (square) or 3 (round)

DESIGN CHECK (run by make_power_pcb.py)
clearance >= 0.6 mm between nets, ground fill 1.0 mm from other nets, all nets connected,
ground path for the Peltier current >= 3.2 mm wide, no part bodies overlapping: OK

HOW IT CONNECTS

